MOHAMMAD KAMRUL HASAN

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Education

Bangladesh University of Engineering and Technology (BUET) Apr 2025 – 2028 (expected)
B.Sc. in Computer Science and Engineering • CGPA 3.77/4.00 *Dhaka, Bangladesh*
Ranked 79th in the national BUET admission merit list

Experience

Backend Developer Intern Jun 2026 – Present
Version3.ai • *conversational-commerce AI for online stores* *Dhaka, Bangladesh*

- Built the multi-tenant product database service in **NestJS**, **MongoDB** and **RabbitMQ**: schema design, tenant provisioning and strict per-store data isolation.
- Designed the **WooCommerce** and **Shopify** store integrations: **OAuth 2.0 PKCE**, HMAC-signed requests, AES-256-GCM secret storage, a durable dispatch queue, and idempotent webhook handling.
- Root-caused a production bug where the OAuth callback returned `201` instead of `200`, which made WooCommerce permanently delete each store's newly issued API keys.
- Benchmarked SigLIP2, Marqo and fashion-tuned CLIP models for visual product search on Products-10K (9,691 SKUs, 132K images) using Recall@k, MRR and nDCG@10 with bootstrap CIs; the chosen model reached **Recall@1 0.672**.
- Showed **Qdrant** int8 and binary quantization cost a median 0.004 Recall@1, and that 4 single-thread workers give **2.2**× the CPU throughput of one 4-thread worker, so search can be served on CPU-only machines.

Software Engineering Intern Dec 2025 – Mar 2026
Edulytics *Dhaka, Bangladesh*

- Built core user flows for an education MVP in **React Native** and **TypeScript** with a remote team, shipping on sprint deadlines.

Projects

LITTLETai — core-aware thread presets for llama.cpp on Android 2026
Arm Create: AI Optimization Challenge (submitted) • POSIX shell, Python, simpleperf, Perfetto • github.com/Kamrulhasan12345/LITTLETai

- Profiled **llama.cpp** on a Helio G81 phone (2×Cortex-A75 + 6×Cortex-A55) with **simpleperf**: the default 8 threads retired up to **5**× more instructions per token, spin-waiting at thread barriers.
- Wrote a launcher that picks cores and thread counts from the CPU topology, without root: **2.2–2.6**× **faster** decode and **2.6–3.3**× **less energy** per token on Qwen2 1B Q4_0.
- Found llama.cpp's CPU-affinity flag was silently ignored on Android (99.5% of cycles stayed on the big cores; since fixed upstream); validated with a reproducible harness (thermal cooldowns, randomized runs, energy logging, 209 test assertions).

Ilham — hadith study platform 2026
Database course project, team of 2 • PostgreSQL, PL/pgSQL, Node.js, Express, TypeScript • github.com/Kamrulhasan12345/ilham

- Designed the **PostgreSQL** schema (ISA hierarchy, weak entities, triggers, audit log) and an ETL that loaded **14,941** hadiths, **139,766** chain positions and 20,957 narrators with **99.6%** narrator resolution.
- Scored each hadith's chain strength in SQL from its weakest link, and locked the reference corpus with `REVOKE` so the application cannot modify it.

Sentinel — offline medical triage app for Android 2025
Lead developer, team of 3 • React Native, TypeScript, TensorFlow Lite • github.com/Kamrulhasan12345/sentinel

- Built an app that sorts symptom descriptions into 52 first-aid intents and Critical/Urgent/Routine **fully offline**: an on-device 1D-CNN (**TF Lite**), falling back to int8 all-MiniLM-L6-v2 embeddings (20 MB) when CNN confidence is under 65%. Released as an APK.

Bayyinah — collaborative desktop Quran app 2026
Java, JavaFX, Spring Boot, WebRTC, SQLite FTS5, PostgreSQL • github.com/Kamrulhasan12345/bayyinah

- Built offline full-text verse search (**SQLite FTS5**) in a JavaFX client, and live group reading sessions with **WebRTC** audio and STOMP signaling on a **Spring Boot** server.

Publication

From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka
UbiComp/ISWC 2026, ACM • co-author (5th of 11), BUET × UC San Diego • I transcribed the interviews behind the qualitative analysis • doi.org/10.1145/3830727.3834827

Skills

Languages: C, C++, Python, TypeScript, JavaScript, Java, SQL (PostgreSQL, PL/pgSQL), POSIX shell
ML & inference: llama.cpp, GGUF quantization, PyTorch, TensorFlow Lite, Hugging Face Transformers, sentence-transformers, SigLIP/CLIP, FAISS, Qdrant
Backend & data: NestJS, FastAPI, Spring Boot, Express, REST, WebSockets, WebRTC, OAuth 2.0, JWT, MongoDB, Redis, RabbitMQ, SQLite
Tools: Linux, Docker, Git, GitHub Actions, simpleperf, Perfetto, adb